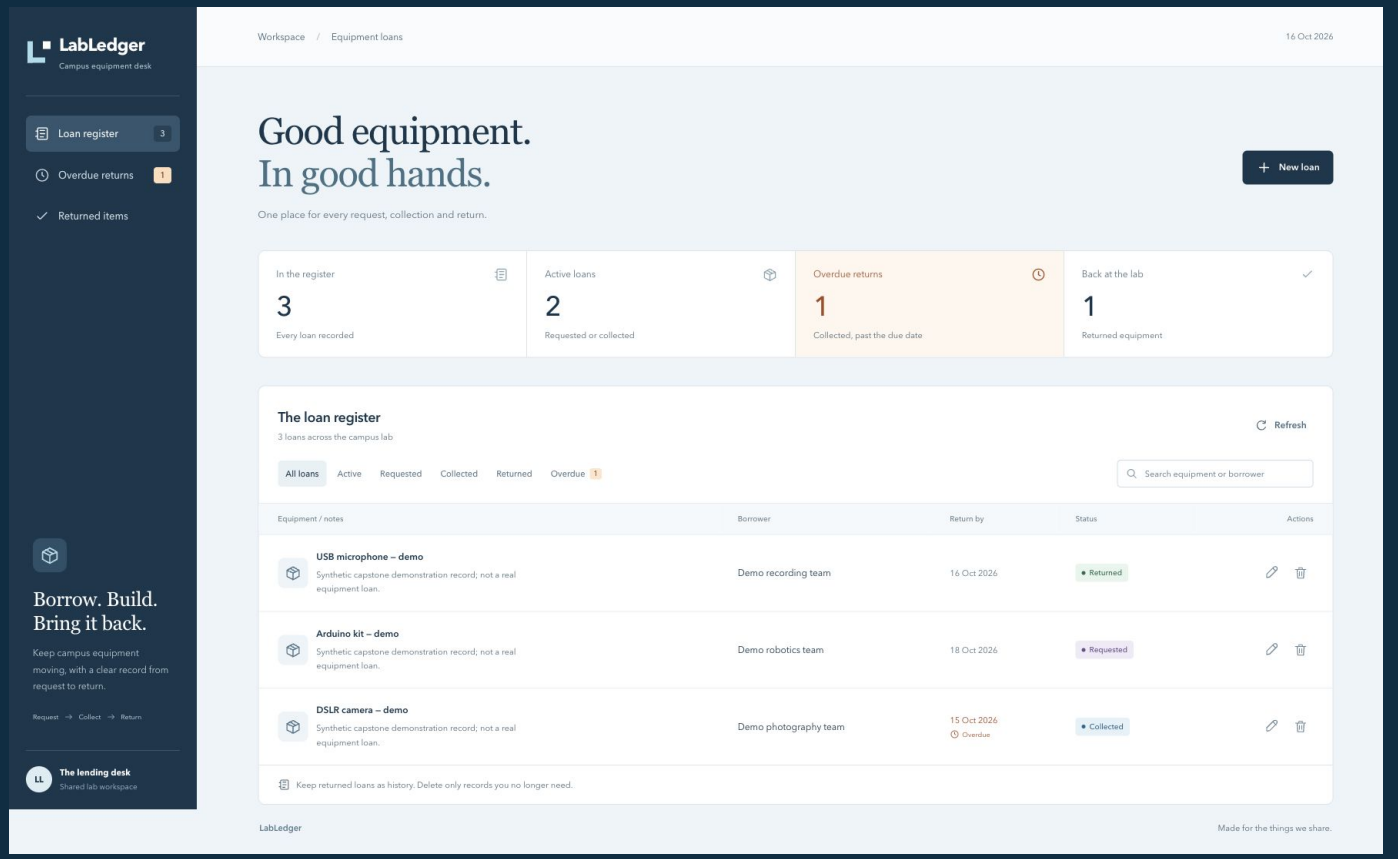
LabLedger

Shared equipment. Clear returns.

React · FastAPI · PostgreSQL
Docker · CI · Helm · Monitoring

VERIFIED ON EKS

Temporary cloud demo cleaned up.



One clear equipment-loan lifecycle

Request → collect → return

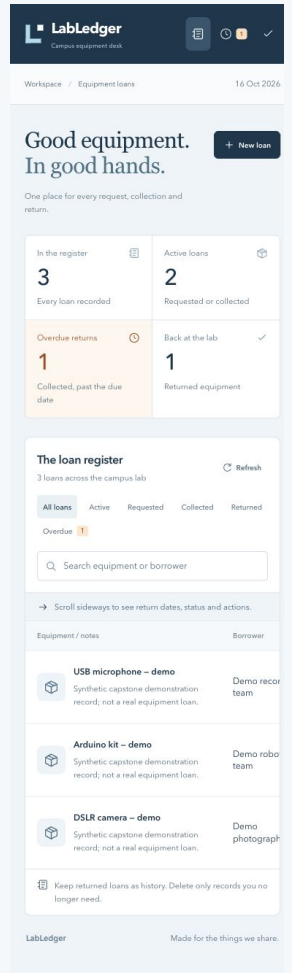
A lab desk records the borrower, equipment and return date.

Overdue has a precise meaning

Collected + due date before today in Asia/Kolkata. Requested loans are active, but not overdue.

Usable on small screens

Search, status filters, readable actions and an accessible dialog. Demonstration records are explicitly fictional.



Genuine School-profile mobile capture

Follow a request through the stack

React UI
Relative /api calls

→

Nginx / Ingress
/ and /api routes

→

FastAPI
Validation + metrics

→

PostgreSQL
Alembic + loans

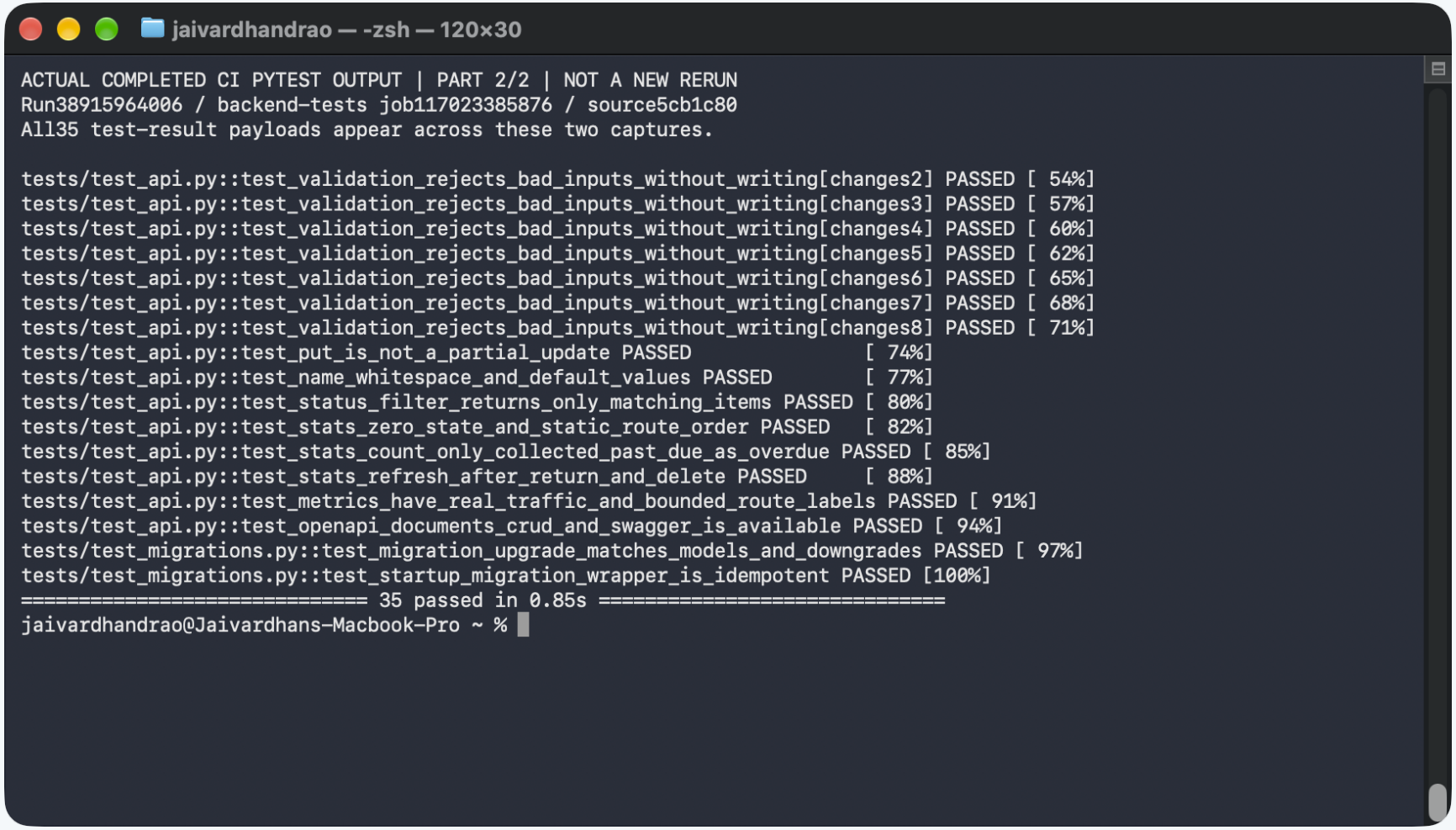
Compose localhost:3000 → frontend → backend → database

Kubernetes localhost:8088 → Ingress → ClusterIP Services → Pods

Prometheus → Grafana

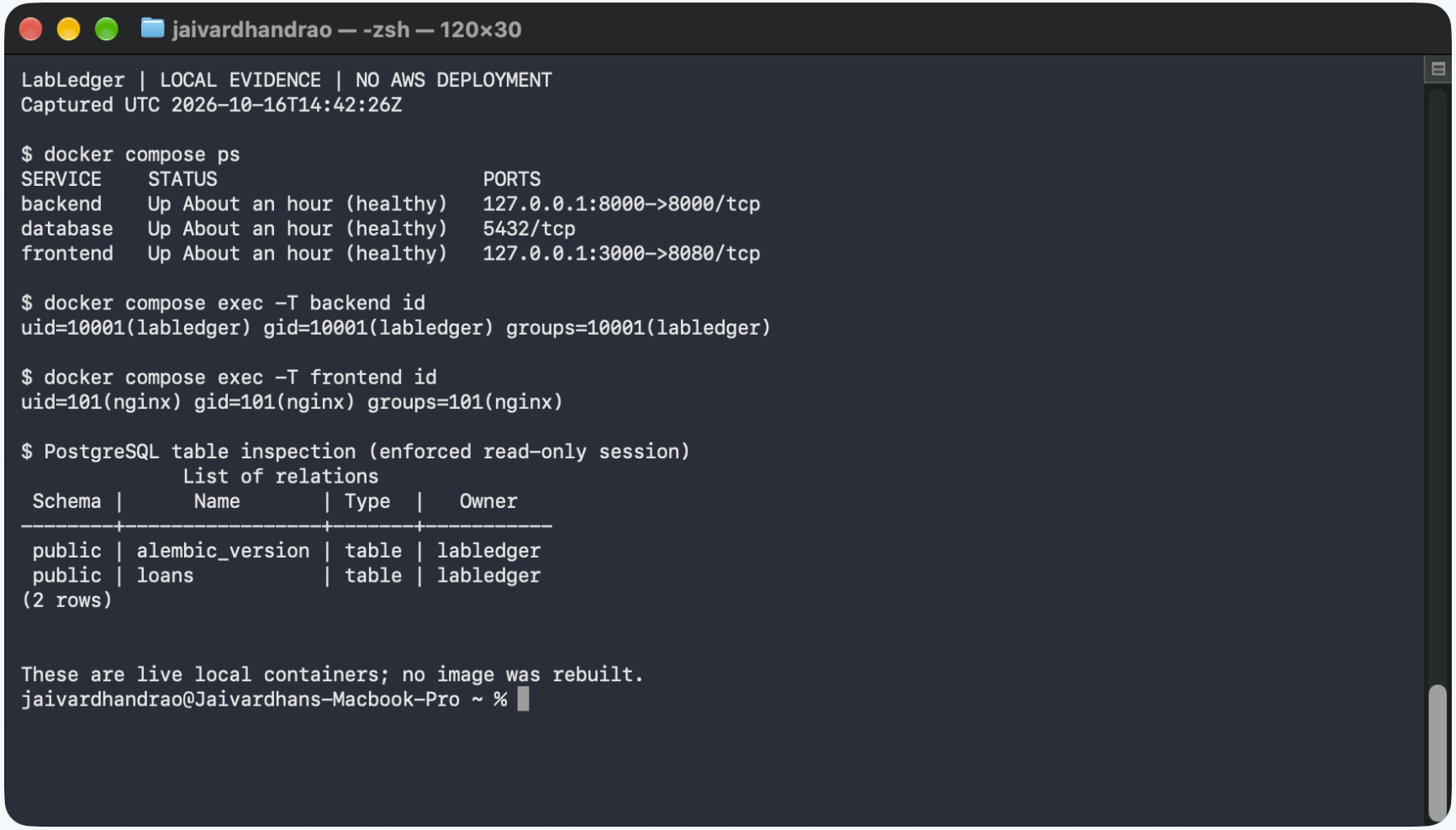
/metrics on both API replicas; real HTTP counters and latency histograms.

35 isolated tests, plus real-stack checks



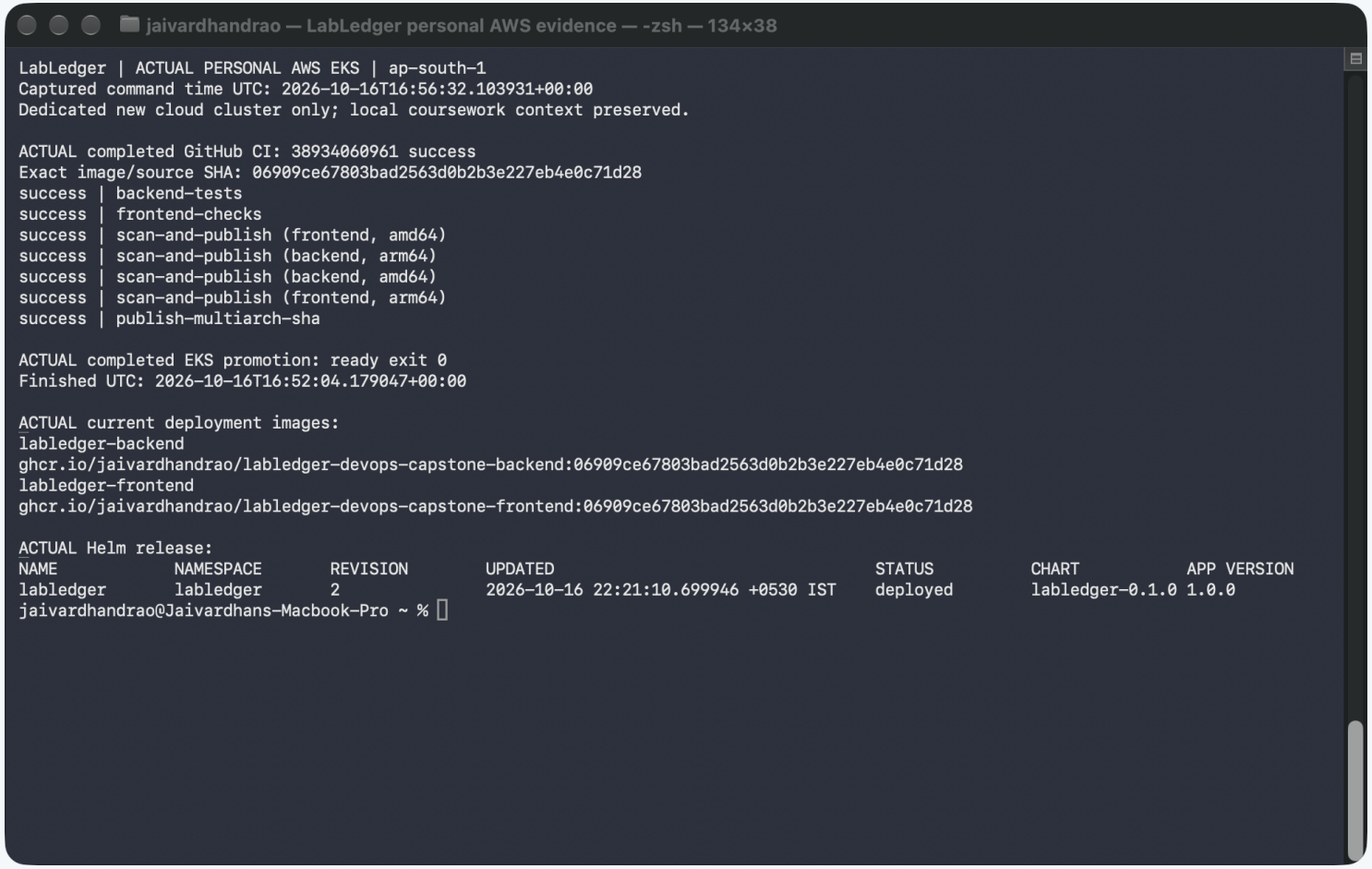
Actual completed CI pytest output; all 35 results are available across the two native captures.

Three healthy services; two non-root images



Live local containers and enforced read-only table inspection; original volumes preserved.

A real change reached the running application



Controller fix → seven green jobs → exact 06909ce EKS rollout, Helm release 2.

Open the actual seven-job green run

The scan gate found a problem, then verified the repair

Before

Strict scan failed

Debian runtime findings were preserved in the original report.

After

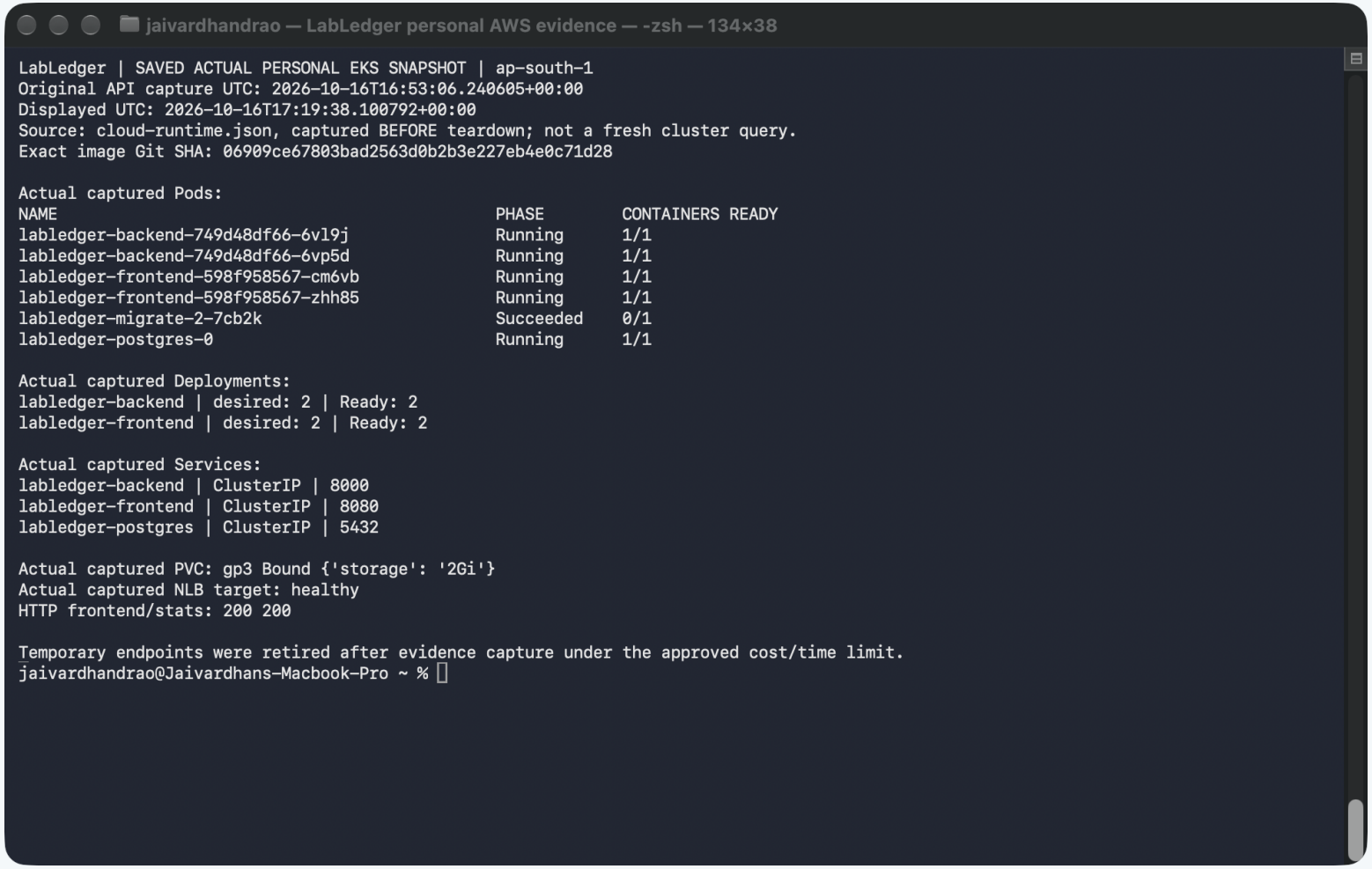
4 strict gates passed

Backend + frontend
amd64 + arm64
No HIGH / CRITICAL allowed

Gate: --severity HIGH,CRITICAL --ignore-unfixed=false --exit-code 1

A clean scan is evidence about that image at that time; it is not a security guarantee.

Helm release 2 ran on the dedicated EKS cluster



Actual EKS: ClusterIP Services, bound gp3, completed migration and deployed Helm releases.

Resilience was exercised, not only configured

2 → 4 Ready → 2

Observed cloud API HPA response; one worker throughout.

7,431 reads · 12 failures

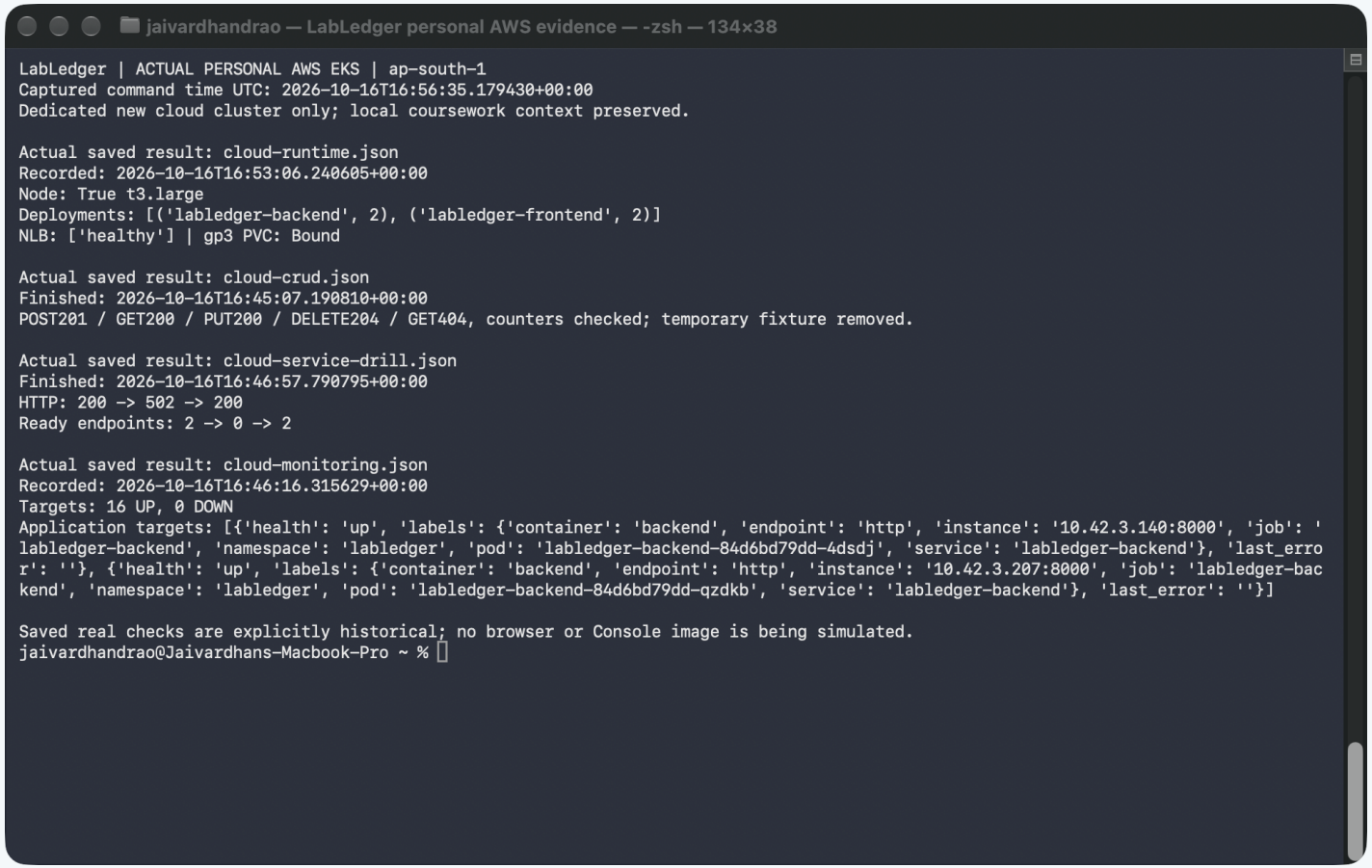
Separate lower-rate recovery: 40/40 HTTP 200. Burst failures retained.

200 → 502 → 200

Service selector fault, diagnosis and real recovery.

Actual cloud records retained. Earlier local 2,338 reads / 0 failures are separate evidence.

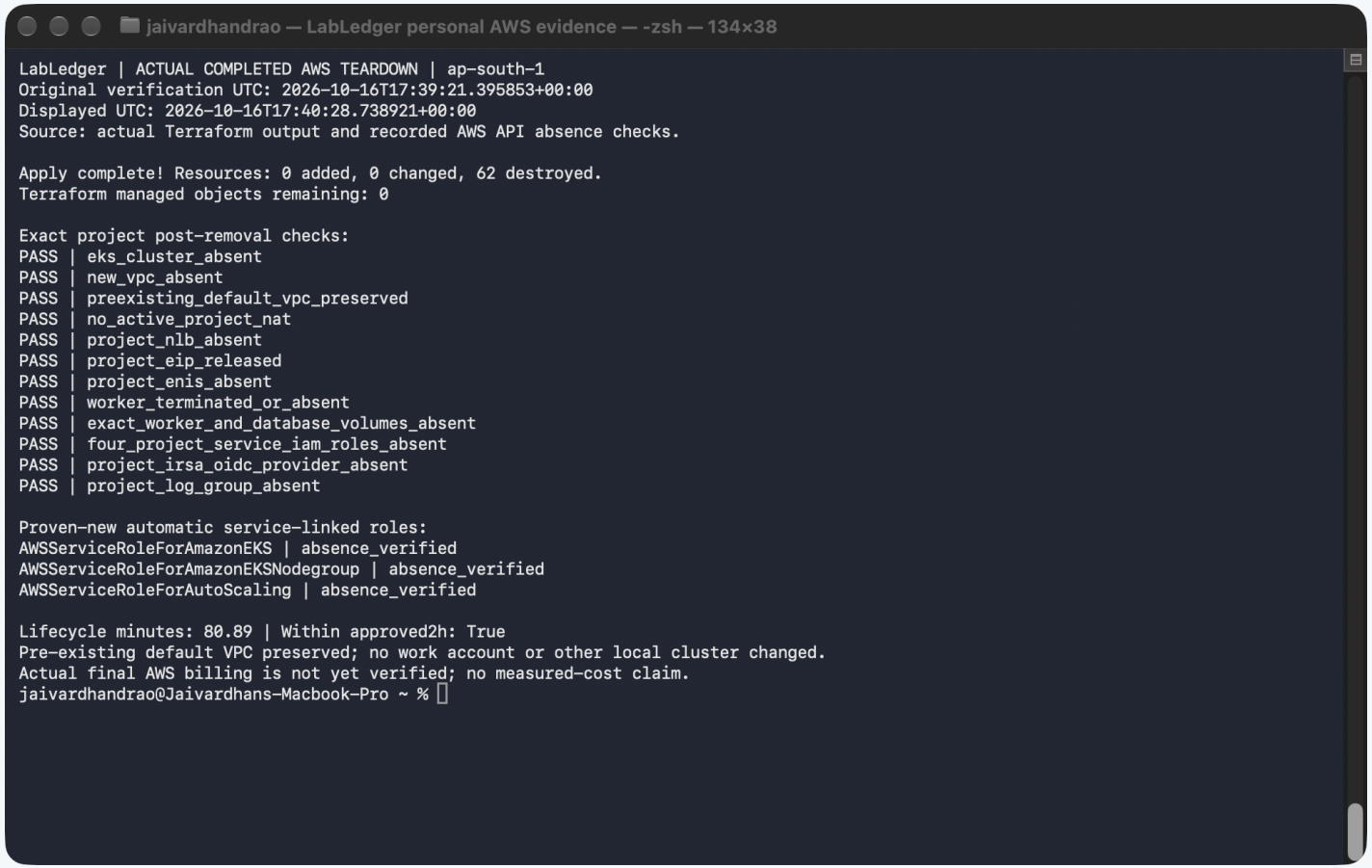
Actual application scraping is healthy



Cloud: 16/16 targets UP at capture time; actual CRUD and Service recovery retained.

Grafana: real API data verified; populated browser-panel capture still pending login.

Real AWS deployment, then verified cleanup



62 created → real runtime verified → 62 destroyed; pre-existing resources preserved.

Planning estimate about US$2.43 before tax; actual final AWS bill is not yet verified.

Present the verified work and its limits

Verified and reproducible

Original app, seven green CI jobs, exact-SHA EKS rollout, real metrics, recovery and verified cleanup.

Remaining evidence limits

Private browser captures need publication approval. Populated Grafana and the full student live walkthrough remain separate.

Continue in the final repository

The student reported manual form submission. No further form action or receipt is claimed.

Public repository and complete evidence index